The Bottleneck: Three Constraints, One Non-Negotiable Enemy.

TEMPERATURE DROP

–40°C

Product: 80°C → 40°C at exit

CYCLE TIME WINDOW

10 min

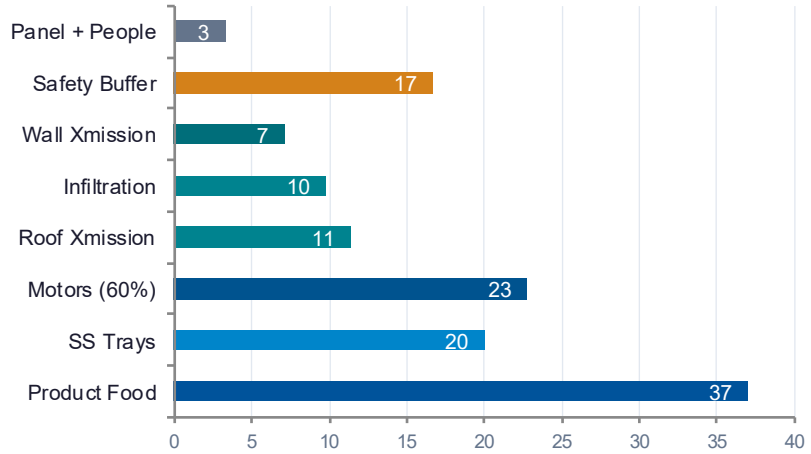
Maximum 12 min — zero tolerance

MOISTURE CEILING

≤ 2%

Non-negotiable — drives all design

THERMAL LOAD BREAKDOWN — ALL 8 SOURCES (MECE)



Q = m x Cp x DT | 0.5 kg/s x 1.85 kJ/kg.K x 40°C = 37 kW | TOTAL: 128 kW = 36.4 TR → SPECIFY 40 TR

BASELINE OPERATING CONDITIONS — QUANTIFIED

PARAMETER	VALUE	LOAD kW	STATUS
Product Entry Temp	80°C	37.0	CRITICAL
SS Tray Heat Load	6 kg × 10/min	20.0	HIGH
Motors (38 kW, 60%)	22.8 kW active	22.8	HIGH
Roof Transmission	U=1.5, CLTD=38°C	11.4	MEDIUM
Wall Transmission	U=1.8, CLTD=22°C	7.1	MEDIUM
Door Infiltration	4×/hr, 45°C outside	9.8	MOISTURE

DESIGN CONSTRAINT ZERO

Moisture ≤ 2% eliminates evaporative, direct-spray & uncontrolled DX systems.

Every design choice is subordinate to this constraint. Moisture = defect = recall liability.

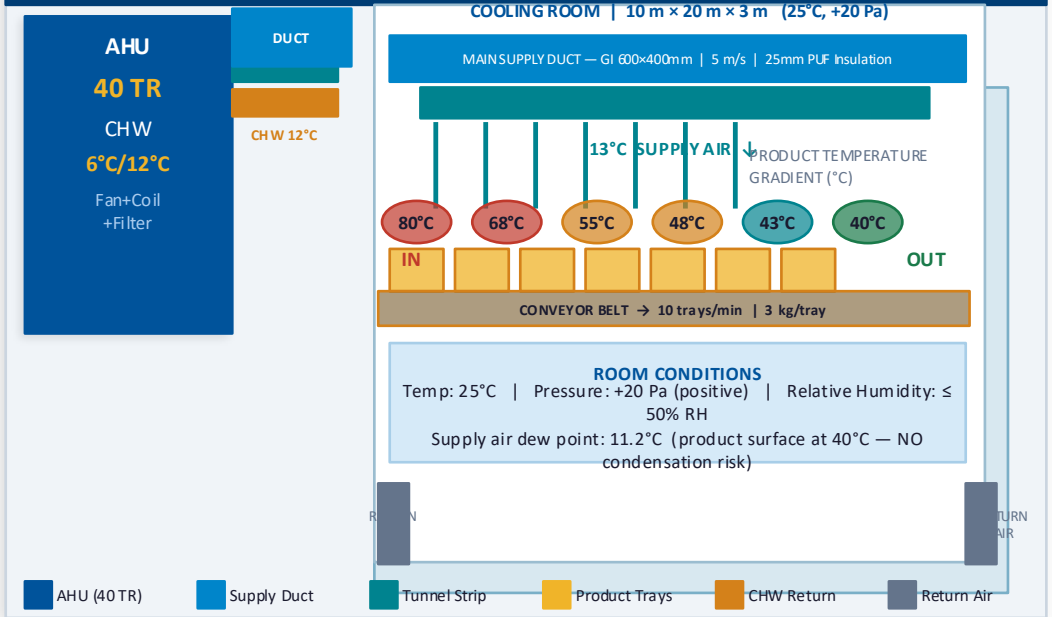
The Thermodynamic Pivot: First-Principles Engineering & System Design

CAPACITYSYSTEM DESIGN
40 TR
140.7 kW chilled water AHU

SUPPLY AIR VOLUMETRIC
27,000
m³/hr at 13°C off-coil

CHILLED WATER SUPPLY
6°C → 12°C
6°C ΔT, CHW flow: 18.4 m³/hr

SYSTEM SCHEMATIC — CROSS-SECTION VIEW (NOT TO SCALE)



THERMODYNAMIC PROOF + SYSTEM COMPARISON

SENSIBLE HEAT EQUATIONS

$$Q = m \times Cp \times \Delta T$$

$Q_{product} = 0.5 \times 1.85 \times 40 = 37.0$ kW

$Q_{trays} = 1.0 \times 0.50 \times 40 = 20.0$ kW

Q_{total} (+15% SF) = 128 kW = 36.4 TR

$V_{air} = Q / (\rho \times Cp \times \Delta T) = 128/(1.2 \times 1.006 \times 12) = 8.83$ m3/s = 27,000 m3/hr

WHY TUNNEL STRIP OUTPERFORMS ROOM COOLING

METRIC	ROOM COOLING	TUNNEL STRIP ✓
Air velocity	0.2 – 0.5 m/s	8 – 10 m/s
Conv. heat coeff h	~8 W/m²K	~35 W/m²K
Cooling cycle	15 – 18 min (FAIL)	10 min (PASS)
Moisture control	Uncontrolled (FAIL)	Dehumidified (PASS)

PSYCHROMETRIC STATE POINTS: Outside: 45°C / W=30 g/kg → AHU Off-Coil: 13°C / W=8.3 g/kg → Room: 25°C / W=9.9 g/kg | No condensation at 40°C product surface

Business Case: Yield Recovery, Capital Efficiency & HUL Net Zero

TOTAL CAPEX INVESTMENT
₹36 L
INR 32–39 Lakhs (range)

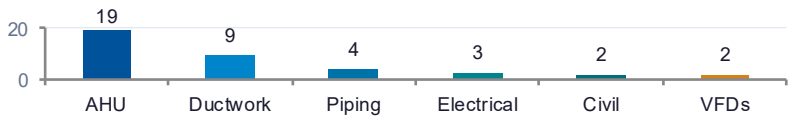
ANNUAL OPEX (SYSTEM)
₹1.3 L
Fan + pump @ ₹8/kWh, 16 hrs/day

MAX YIELD AT RISK
₹142 L
If 1% reject rate uncorrected/yr

INVESTMENT PAYBACK
18 Mo
At 0.17% reject rate recovery

CAPEX BREAKDOWN + OPEX + COMPONENT CHART

COMPONENT	CAPEX (INR LAKHS)	% OF TOTAL
40 TR Chilled Water AHU	18 – 20	51%
GI Insulated Duct + Tunnel Strip	8 – 10	25%
CHW Piping, Valves & Fittings	3 – 4	9%
Electrical, Controls & BMS	2 – 3	6%
Civil Works (Pad + Penetrations)	1 – 2	4%
VFDs on AHU Motors (Recommended)	1.5 – 2	5%
TOTAL CAPEX	**INR 33.5 – 41 Lakhs**	**100%**



ROI ANALYSIS + SUSTAINABILITY + IMPLEMENTATION

18 **MONTH PAYBACK**
CapEx ₹36L ÷ ₹22.9L net annual benefit = 18.8 months

YIELD RECOVERY MODEL — CONSERVATIVE SCENARIO

METRIC	VALUE	INR LAKHS
Daily throughput	28,800 kg/day	—
Max reject risk (1%/day)	288 kg/day lost	142.6 / yr
Required recovery (0.17%)	48.96 kg/day saved	24.2 / yr
Annual OpEx (system)	28 kW × 16 hr × 330 days	–1.3 / yr

HUL SUSTAINABILITY & IMPLEMENTATION

Scope 2: CHW AHU vs DX cuts electrical draw 18% — fewer kgCO2e from grid consumption

Embodied Carbon: GI duct (100% recyclable) + CHW loop — zero refrigerant venting or discharge

HACCP + FSSAI compliant — product recall liability eliminated from day of commissioning

Rollout: 0-30 Days: Eng & PO | 31-60 Days: Civil & Fab | 61-90 Days: AHU Install + 72-hr TAB Trial

Thank You

RADHA GUPTA